COMMUTATIVITY OF DERIVED GLOBAL SECTIONS WITH THE DERIVED PUSHFORWARD AND THE RESTRICTION OF SCALARS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement ends by pointing the reader to a remark for more explanation; the remark is carried as a navigation stub and is not part of the proof, which is complete as written.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-before-Leray is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Right derived functor of an exact functor is the functor itself. Derived Categories, Lemma derived-lemma-right-derived-exact-functor, invoked at proof lines 2125-2128. Supplies that the derived restriction functor is computed by applying the functor.
- Composition of derived functors and the acyclicity condition. Derived Categories, Lemma derived-lemma-compose-derived-functors, invoked at proof lines 2137-2139. Supplies the commutativity once the hypotheses are verified for both compositions.
- Pushforward of an injective module is acyclic for global sections. Same chapter, Lemma lemma-pushforward-injective, invoked at proof lines 2141-2143. Supplies the acyclicity hypothesis for the second composition.

Source reference. Original lemma 015M: sources/stacks-chow/tags.txt, source lines 1513–1513. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 015F: sources/stacks-chow/tags.txt, source lines 1506–1506. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EX: sources/stacks-cohomology/pinned-original-source.tex, source lines 1779–1779. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original remark 01F0: sources/stacks-cohomology/pinned-original-source.tex, source lines 2147–2147. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. There is a commutative diagram*

$$\begin{array}{ccc} D^+(X) & \xrightarrow{R\Gamma(X,-)} & D^+(\mathcal{O}_X(X)) \\ {\scriptstyle Rf_*}\downarrow & & \downarrow{\scriptstyle restriction} \\ D^+(Y) & \xrightarrow{R\Gamma(Y,-)} & D^+(\mathcal{O}_Y(Y)) \end{array}$$

More generally for any $V \subset Y$ open and $U = f^{-1}(V)$ there is a commutative diagram

$$\begin{array}{ccc} D^+(X) & \xrightarrow{R\Gamma(U,-)} & D^+(\mathcal{O}_X(U)) \\ {\scriptstyle Rf_*}\downarrow & & \downarrow{\scriptstyle restriction} \\ D^+(Y) & \xrightarrow{R\Gamma(V,-)} & D^+(\mathcal{O}_Y(V)) \end{array}$$

See also Remark 01F0 for more explanation.

Proof. Let $\Gamma_{res} : Mod(\mathcal{O}_X) \to \mathrm{Mod}_{\mathcal{O}_Y(Y)}$ be the functor which associates to an $\mathcal{O}_X$-module $\mathcal{F}$ the global sections of $\mathcal{F}$ viewed as an $\mathcal{O}_Y(Y)$-module via the map $f^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$. Let $restriction : \mathrm{Mod}_{\mathcal{O}_X(X)} \to \mathrm{Mod}_{\mathcal{O}_Y(Y)}$ be the restriction functor induced by $f^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$. Note that *restriction* is exact so that its right derived functor is computed by simply applying the restriction functor, see Derived Categories, Lemma 015F. It is clear that

$$\Gamma_{res} = restriction \circ \Gamma(X, -) = \Gamma(Y, -) \circ f_*$$

We claim that Derived Categories, Lemma 015M applies to both compositions. For the first this is clear by our remarks above. For the second, it follows from Lemma 01EX which implies that injective $\mathcal{O}_X$-modules are mapped to $\Gamma(Y, -)$-acyclic sheaves on Y. □